Frames and matrix designs

Universal exponential deterioration of minimally redundant real phase-retrieval frames

Difficulty: challenging **Importance:** interesting to the community

Status: Open

Rating rationale: A universal bound over every full-spark measurement design requires a structural obstruction beyond the Gaussian analysis; it matters for stable phase retrieval.

For $n \geq 2$, let $A \in \mathbb{R}^{(2n-1)\times n}$ be full spark: every choice of n rows is linearly independent. Write a_i^T for its rows and define

$$L(A) = \max_{1\leq i\leq 2n-1} \|a_i\|_2, \qquad \omega(A) = \min_{\substack{T\subseteq\{1,\dots,2n-1\}\\ |T|=n}} \sigma_{\min}(A_T).$$

Do absolute constants $C > 0$ and $0 < \beta < 1$ exist such that

$$\omega(A) \leq CL(A)\beta^n$$

for every $n \geq 2$ and every such A?

For a full-spark matrix at this row count, the displayed definition is equivalent to the Balan–Wang definition taking the minimum over row sets whose complements fail to span $\mathbb{R}^n$. It measures a worst-conditioned square subproblem in inversion of the phaseless map $x \mapsto |Ax|$. The conjecture says that clever deterministic measurement design cannot avoid exponential deterioration at minimal real redundancy.

References

1. R. Balan and Y. Wang, *Invertibility and robustness of phaseless reconstruction*, Applied and Computational Harmonic Analysis 38 (2015), pp. 469–488. Conjecture 5.1 on printed p. 484; definitions and stability comparisons in Sections 3–5. Author-hosted paper.
2. A. S. Bandeira et al., *Randomstrasse 101: Open Problems of 2025*, arXiv:2603.29571 (2026), Conjecture 20. Paper.
3. Y. Shmalo, *Extreme least singular values of Gaussian row submatrices and a phase retrieval stability problem*, arXiv:2607.06249 (2026). Abstract and Section 1.2/Corollary 1.2 give the critical real Gaussian asymptotic. Paper.

Status check — 2026-09-10

Searched "Balan Wang exponential conjecture solved", "phase retrieval stability omega 2026", and checked the latest abstract of arXiv:2607.06249. The July 2026 result proves $\omega(A_n) = 4^{-n+o_P(n)}$ for independent standard Gaussian entries. A probabilistic result for this ensemble does not prove the uniform inequality over all full-spark matrices. The separate Gaussian-base question should therefore be excluded, while this deterministic conjecture remains open in the screened literature.

Audit update (2026-09-10): Rechecked Conjecture 20 and Shmalo's July 2026 Gaussian asymptotic, then searched for universal Balan–Wang bounds. A high-probability ensemble theorem is not a deterministic bound for every full-spark matrix, so the target remains open. This is a bounded literature check, not a proof that no solution exists.